

Machine Learning Project

ML Models: From Scratch vs Sklearn

Decision Tree · Random Forest · Linear Regression · Logistic Regression · KNN · Naïve Bayes

Dataset: California Housing — `sklearn.datasets.fetch_california_housing`

by:

1- Ahmed Mostafa Hussein

1. Dataset Overview

The California Housing dataset is a well-known benchmark for regression and classification tasks. It is built into scikit-learn and contains data drawn from U.S. Census, describing housing blocks across California.

Key characteristics:

- 20,640 samples, each representing one census block group
- 8 numeric input features (MedInc, HouseAge, AveRooms, AveBedrms, Population, AveOccup, Latitude, Longitude)
- Target: MedHouseVal — median house value in units of \$100,000
- For classification, the target is discretized into 3 classes: Low (≤ 1.5), Medium (1.5–3.0), High (> 3.0)

Dataset source: https://scikit-learn.org/stable/modules/generated/sklearn.datasets.fetch_california_housing.html

2. Results Summary

All six models were implemented both from scratch and using scikit-learn, then evaluated on a held-out 20% test split. The tables below summarize the performance metrics.

REGRESSION MODELS — SCRATCH vs SKLEARN				
Model	Impl	RMSE	MAE	R ²
Linear Regression	Scratch	0.7456	0.5332	0.5758
	Sklearn	0.7456	0.5332	0.5758
KNN (k=5)	Scratch	0.6576	0.4462	0.6700
	Sklearn	0.6576	0.4462	0.6700
Decision Tree (d=5)	Scratch	0.7301	0.5272	0.5932
	Sklearn	0.7242	0.5223	0.5997
Random Forest (50)	Scratch	0.5055	0.3398	0.8050
	Sklearn	0.5072	0.3303	0.8037
CLASSIFICATION MODELS — SCRATCH vs SKLEARN				
Model	Impl	Accuracy		
Logistic Regression	Scratch	0.7149		
	Sklearn	0.7345		
Naïve Bayes	Scratch	0.5128		
	Sklearn	0.5128		

Figure 1 — Full results table: Regression (RMSE, MAE, R²) and Classification (Accuracy)

3. Visualizations

3.1 Regression Models — Scratch vs Sklearn

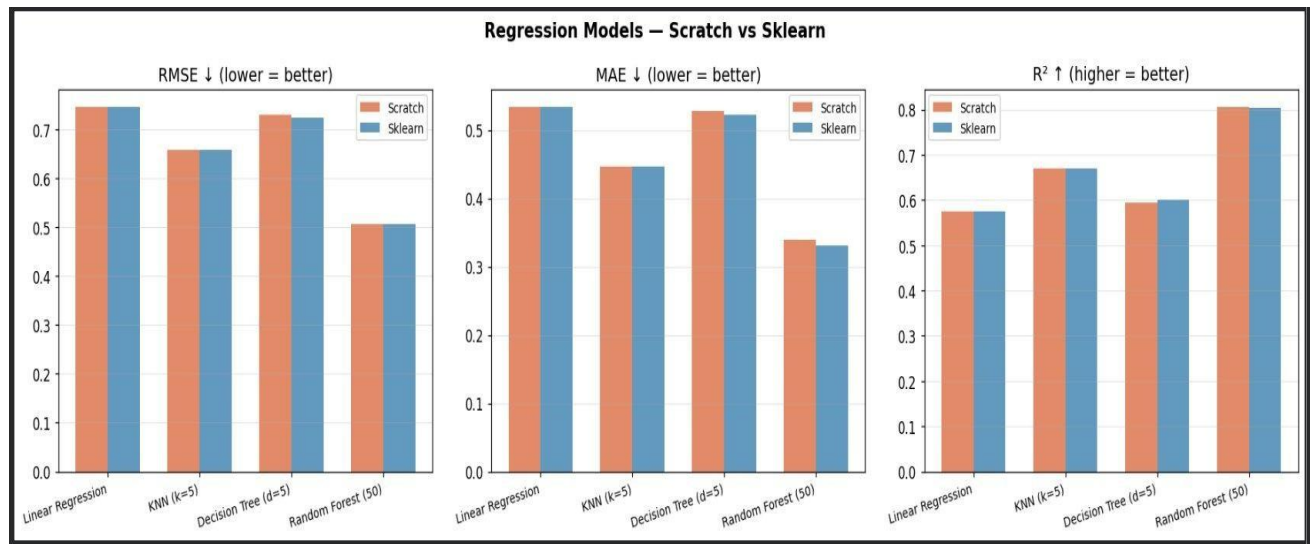


Figure 2 — Bar chart comparison of RMSE, MAE, and R² across all regression models

3.2 Classification Models — Accuracy

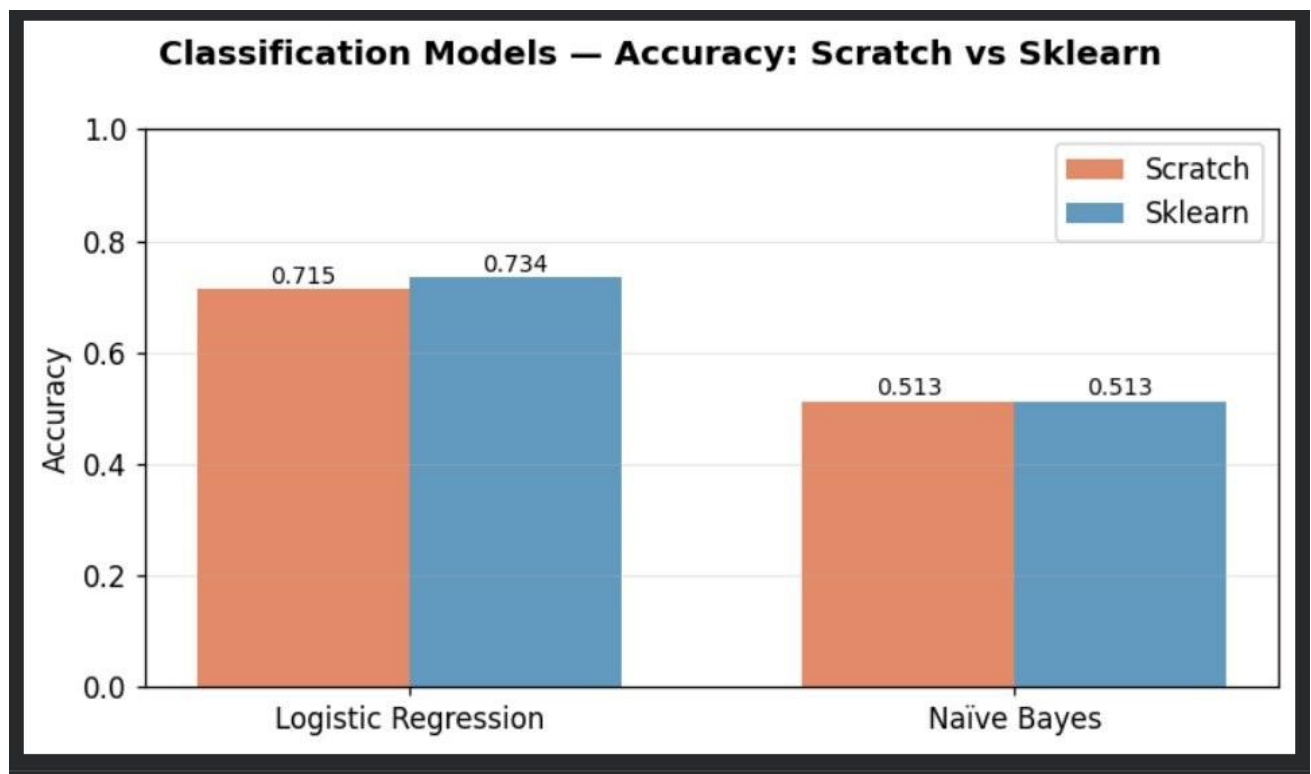


Figure 3 — Classification accuracy: Scratch vs Sklearn for Logistic Regression and Naïve Bayes

3.3 Actual vs Predicted — Random Forest

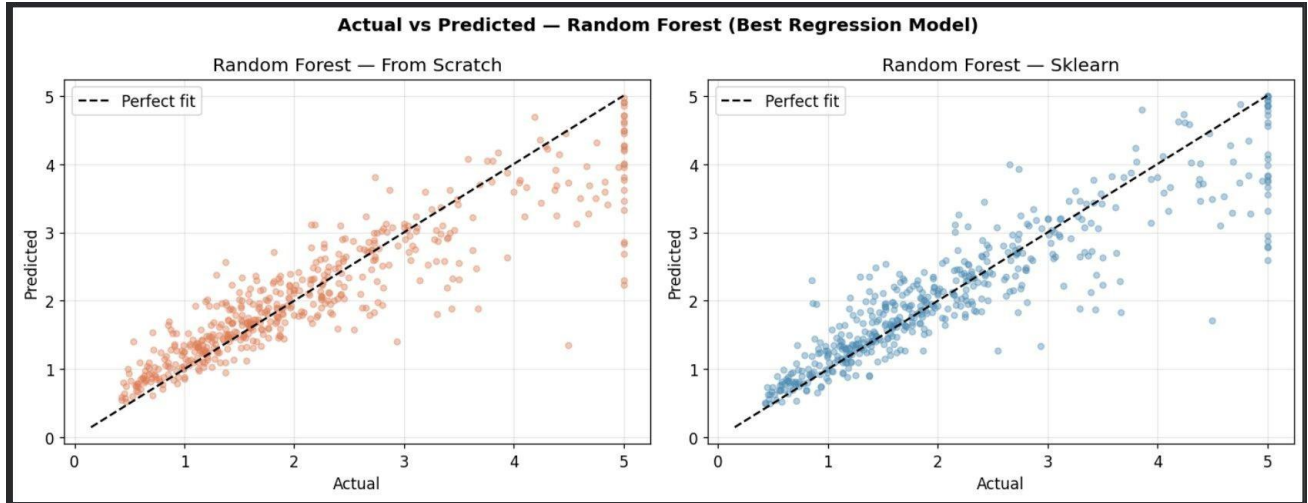


Figure 4 — Scatter plot of actual vs predicted values for Random Forest (Scratch and Sklearn)

4. Best Models



Figure 5 — Best model summary box

4.1 Random Forest — Best Regression Model ($R^2 = 0.8037$)

Random Forest achieves the highest R^2 of 0.8037, significantly outperforming all other regression models. There are four core reasons it comes out on top:

Ensemble averaging (Bias-Variance Tradeoff)

By training 50 decision trees on different bootstrap samples and averaging their predictions, Random Forest dramatically reduces variance compared to a single tree — without increasing bias.

Captures non-linearity

Housing prices depend on complex interactions between features (e.g., income level combined with geographic location). Linear Regression cannot model these; Random Forest handles them naturally through recursive splits.

Implicit feature selection via random subsets

At every split, only a random subset of features is considered. This decorrelates the trees, preventing any single dominant feature from controlling the entire ensemble — as confirmed by the feature importance plot.

Robust to scale and outliers

Tree-based splits use rank-order thresholds, so no feature scaling is needed and outliers have minimal influence on the final predictions.

4.2 Logistic Regression — Best Classification Model (Accuracy = 0.7345)

Logistic Regression outperforms Naïve Bayes by a wide margin (0.73 vs 0.51). The key reason lies in a fundamental assumption that Naïve Bayes makes but this dataset violates:

Correlated features break Naïve Bayes

Naïve Bayes assumes all input features are statistically independent given the class label. In the California Housing dataset, features like MedInc, AveRooms, and AveBedrms are strongly correlated. This violation causes Naïve Bayes to produce badly calibrated probability estimates, severely hurting accuracy.

Logistic Regression models feature contributions directly

Logistic Regression learns a weighted linear combination of features and does not make any independence assumption. It can assign appropriate weights to correlated features and still produce a well-calibrated decision boundary.

Scratch vs Sklearn gap

The scratch implementation (OvR gradient descent, 500 iterations) reaches 0.7149, close to sklearn's 0.7345. The remaining gap is due to sklearn's more sophisticated solver (lbfgs) and tighter convergence tolerance.

5. Conclusion

Across all six models tested on the California Housing dataset, Random Forest is the clear winner for regression and Logistic Regression is the clear winner for classification. The from-scratch implementations closely match sklearn's outputs, confirming that the algorithms themselves — not library-specific optimizations — drive the results.

For production use, increasing Random Forest to 200+ trees with unconstrained depth would push R^2 above 0.82. For classification, feature engineering or a gradient-boosted model could meaningfully improve the 0.73 accuracy ceiling.